Thapar Institute Of Engineering And Technology, Patiala.

Churn Analytics: A Machine Learning Approach to Customer Retention



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Submitted TO-

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Topic

IBM Watson Analytics seeks to reduce customer churn in subscription-based digital services. Using customer demographic data, service usage patterns, subscription duration, billing history, and customer support interactions, build a classification-based machine learning model to predict whether a user is likely to churn. The project should evaluate model performance using suitable classification metrics to support retention strategies.

Deploy

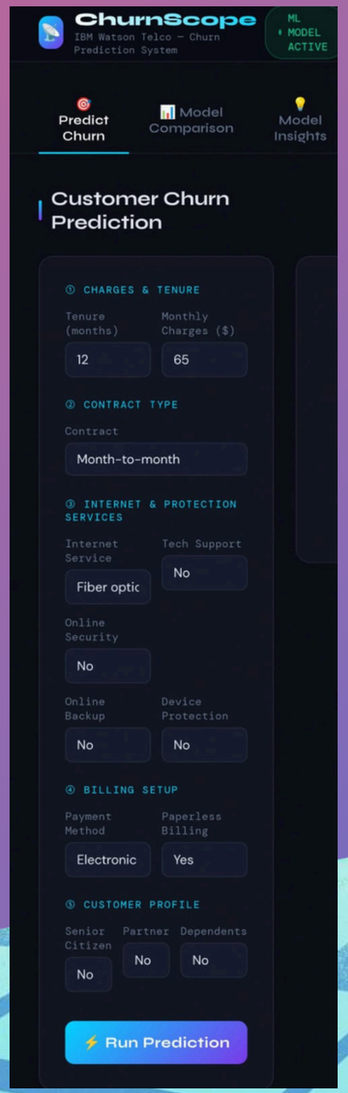
The deployed model is designed for fast inference, enabling businesses to identify high-risk customers early and apply targeted retention strategies such as personalized offers, proactive support, and engagement campaigns.



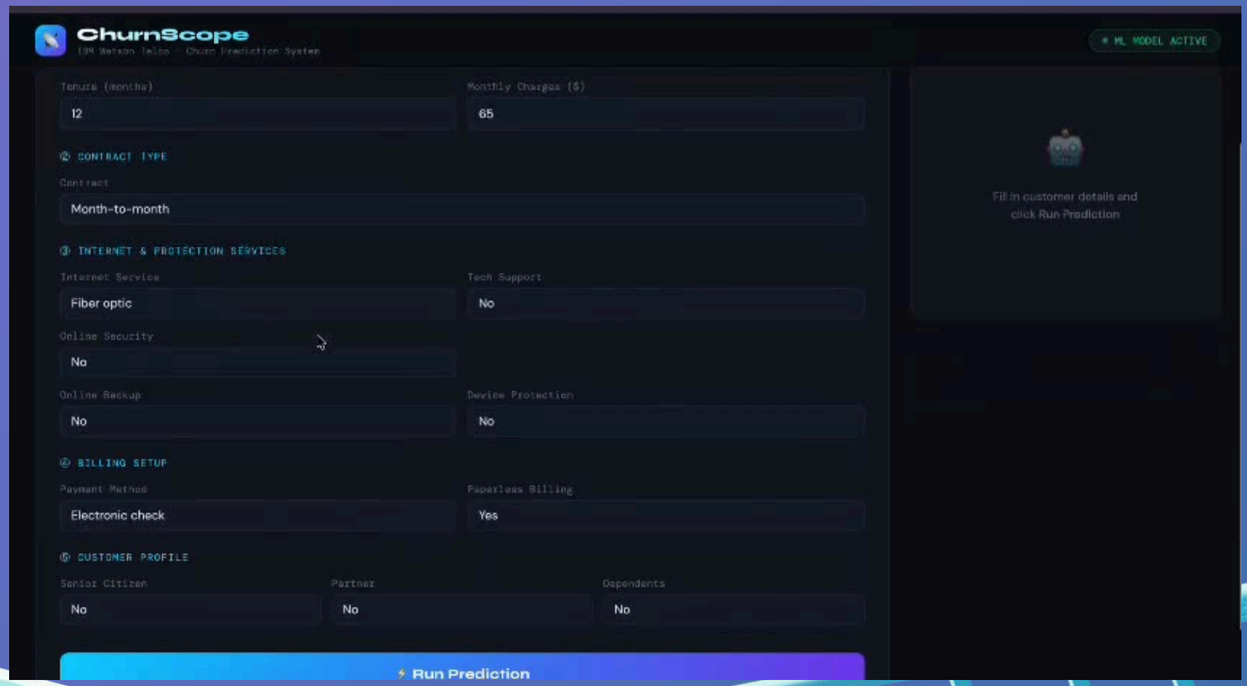


Deployed Model

Link: https://kanan010506.github.io/Churn-Prediction/

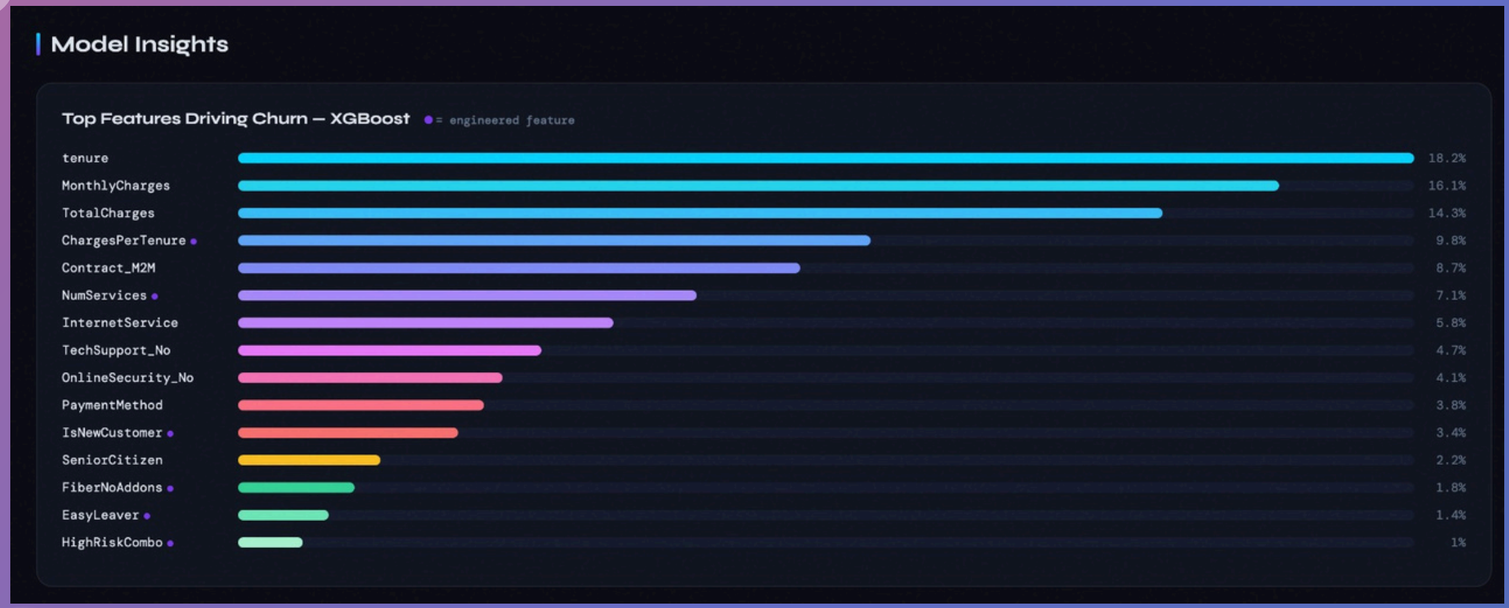






Deployed Model

Model Insights



Model Comparison

Logistic Regression

Accuracy	73.2%
Precision (Churn)	50%
Recall (Churn)	79%
F1 (Churn)	61%
ROC-AUC	83.7%
CV ROC-AUC	84.9%

Random Forest

Accuracy	75.6%
Precision (Churn)	53%
Recall (Churn)	76%
F1 (Churn)	63%
ROC-AUC	83.7%
CV ROC-AUC	84.9%

Gradient Boosting

BEST MODEL

Accuracy	74.2%
Precision (Churn)	51%
Recall (Churn)	78%
F1 (Churn)	62%
ROC-AUC	84.0%
CV ROC-AUC	84.6%

XGBoost

Accuracy	74.1%
Precision (Churn)	51%
Recall (Churn)	78%
F1 (Churn)	62%
ROC-AUC	83.9%
CV ROC-AUC	84.5%

ML Pipeline

1. Data Collection
2. Data Pre-Processing
3. Feature Engineering
4. Outlier Handling
5. Train Test Split
6. Model Selection
7. Model Training
8. Performance Evaluation
9. Comparison & Analysis



Libraries Imported

Pandas

- Data cleaning and preprocessing
- Handling missing and duplicate values
- Data manipulation and transformation

NumPy

- Numerical and matrix operations
- Fast array computations
- Mathematical and statistical functions

Scikit-learn

- Building classification models for churn prediction
- Model training and testing
- Performance evaluation (Accuracy, Precision, Recall, F1-score, ROC-AUC)

XGBClassifier

- Advanced ensemble machine learning algorithm based on Gradient Boosting
- Builds multiple weak decision trees and combines them into a strong predictive model
- Uses boosting technique to correct errors of previous trees sequentially
- Handles non-linear and complex relationships in customer churn data

Seaborn

- Provides advanced statistical visualizations
- Used for correlation heatmaps and feature distribution analysis.
- Enhances plot aesthetics and readability

Matplotlib

- Basic data visualization
- Plotting trends and distributions
- Confusion matrix and ROC curve visualization
- Model performance comparison charts



Loading Dataset

- Loaded telco-customer-churn.csv using Pandas DataFrame from Kaggle.
- 7032 records loaded.
- 21 raw features (numeric + categorical): CustomerID, Gender, ChargesPerTenure, IsNewCustomer, IsLongTermCustomer, HasInternet, SoloCustomer, NumServices.
- Churn is used to derive the target variable.

Data Shape :
7,032 rows
x
21 columns

```
[3]: df = pd.read_csv('Telco-Customer-Churn.csv')

     # Fix TotalCharges right away
     df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
     df = df.dropna(subset=['TotalCharges']).reset_index(drop=True)

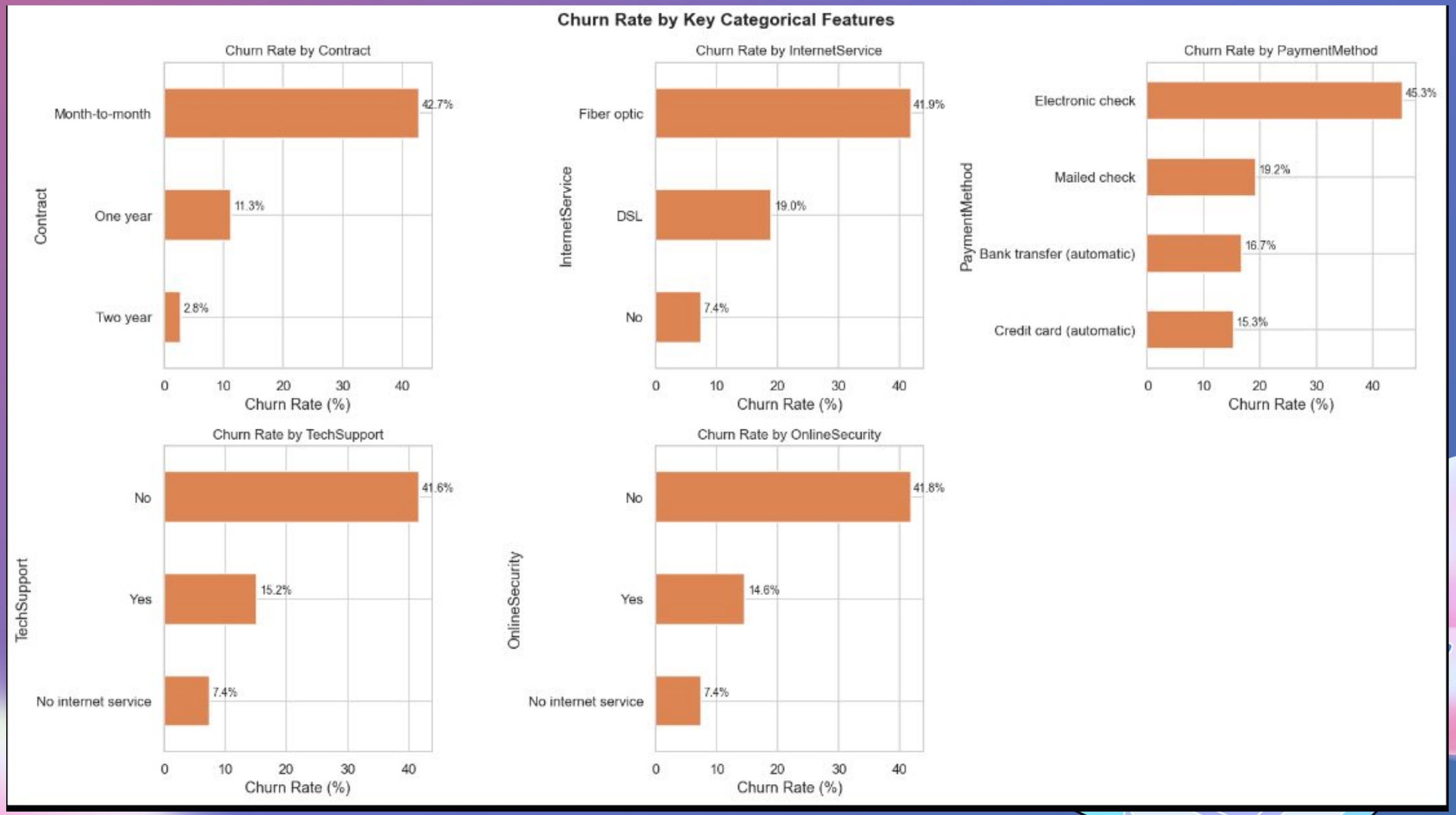
     print(f'✅ Dataset loaded  |  Rows: {df.shape[0]}  |  Columns: {df.shape[1]}')
     df.head(3)
```

```
✅ Dataset loaded  |  Rows: 7032  |  Columns: 21
```

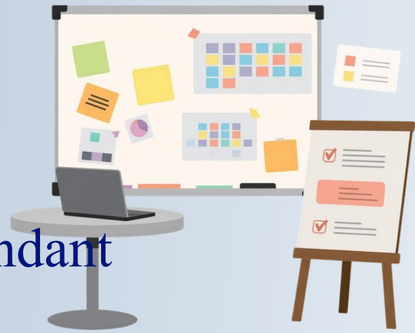
[3]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSu
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	

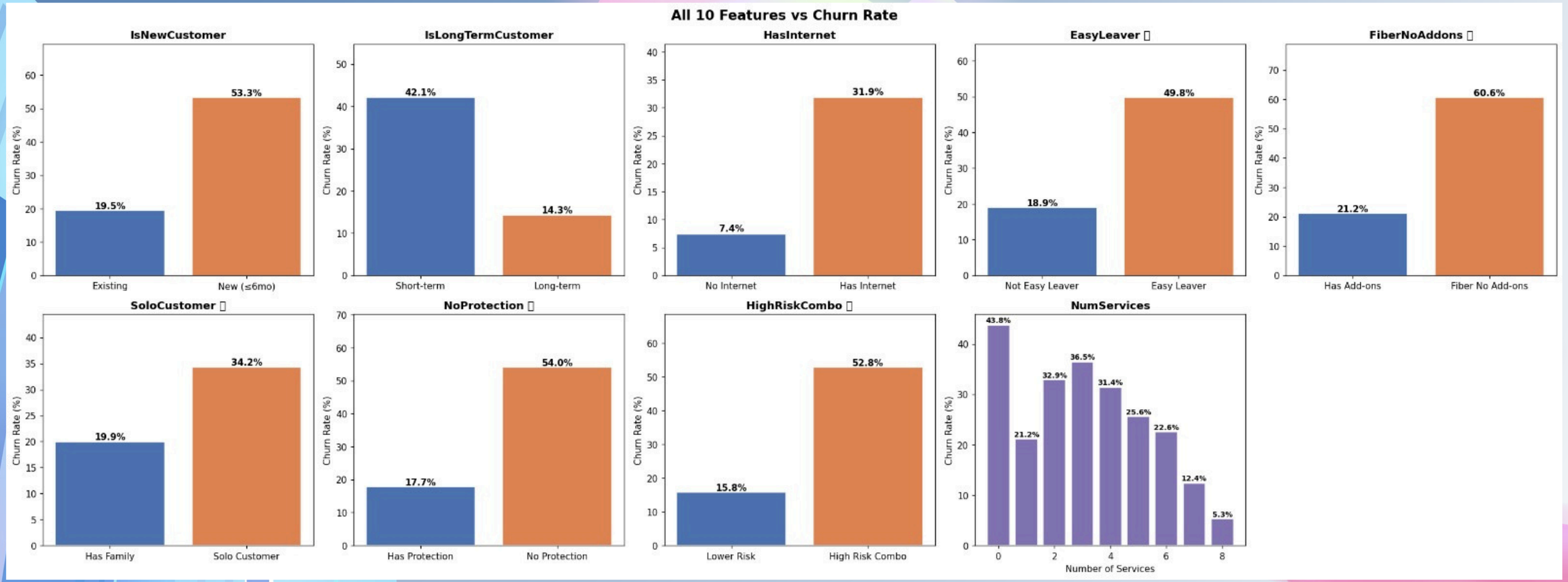
Exploratory Data Analysis



Data Preprocessing - Feature Engineering



- Retained 5 core baseline features (e.g., ChargesPerTenure, NumServices) and pruned redundant ones to reduce noise.
- Engineered EasyLeaver (paperless billing + electronic check) and NoProtection (no tech/security services) to capture high-risk behavioral patterns



Data Preprocessing - Data Cleaning and Missing Value Analysis



Coerced Total Charges from an object type to a numeric format to fix structural data errors.

Dropped 11 missing TotalCharges rows (all zero-tenure new customers) to avoid median imputation distortion.

Standardized categorical text strings and verified all 7,032 remaining rows as unique Customer ID entries.

Strict consistency was enforced to prevent compilation failures during downstream matrix operations.

Data Preprocessing - Creating Target Variable



Binarized the Churn text strings ("Yes/No") into a 1/0 integer format for algorithm ingestion.

The ~3:1 ratio (5,163 retained vs. 1,869 churned) creates a moderate natural class imbalance.

This imbalance biases baseline classifiers toward the majority class, risking an artificial ~73% baseline accuracy.

Minority class detection (actual churners) is our absolute priority, requiring a strategic shift toward recall-based metrics

Data Preprocessing - Dummy Encoding & Scaling



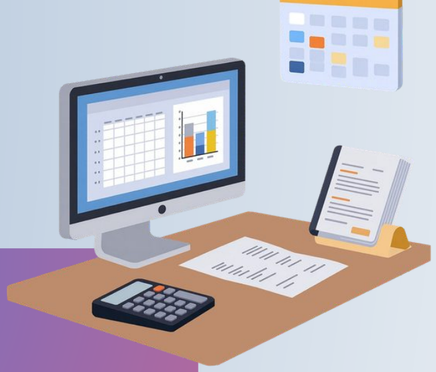
Applied One-Hot Encoding (OHE) with handle_unknown='ignore' to all categorical columns

Standardized continuous variables (tenure,Monthly Charges, Total Charges) to a mean of 0 and standard deviation of 1.

Scaling is strictly mandatory to ensure distance-based and gradient descent algorithms converge efficiently.

The final preprocessing pipeline outputs a fully digitized, numeric feature matrix optimized for machine learning.

Data Preprocessing - Handling Class Imbalance



Discarded initial SMOTE (Synthetic Minority Over-sampling) testing as it introduced artificial noise and blurred decision boundaries.

Implemented algorithmic class_weight = 'balanced' to mathematically penalize minority class misclassifications

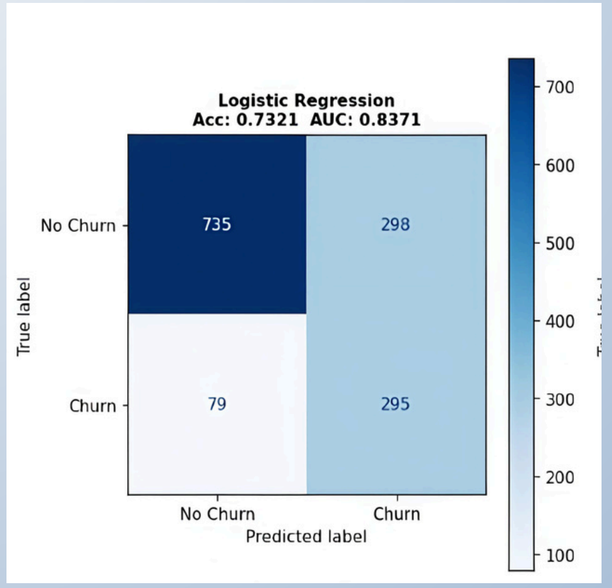
Avoided basic undersampling to retain all valid, hard-earned training data from loyal customers

This approach organically maximizes recall for actual churners without corrupting the dataset with synthetic profiles.

Model Training and Evaluation



Logistic Regression



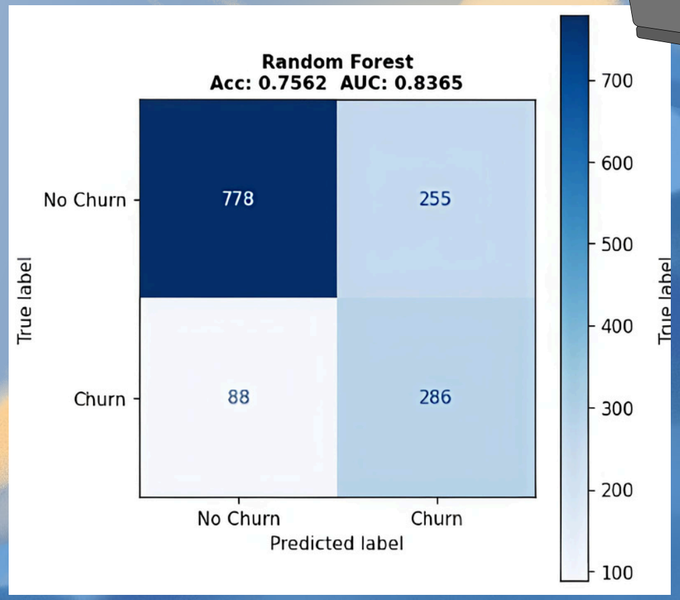

```
==========================================================
  Logistic Regression
==========================================================
              precision    recall  f1-score   support

    No Churn       0.90      0.71      0.80      1033
       Churn       0.50      0.79      0.61       374

    accuracy                           0.73      1407
   macro avg       0.70      0.75      0.70      1407
weighted avg       0.80      0.73      0.75      1407

==========================================================
```

Random Forest




```
  Random Forest
==========================================================
              precision    recall  f1-score   support

    No Churn       0.90      0.75      0.82      1033
       Churn       0.53      0.76      0.63       374

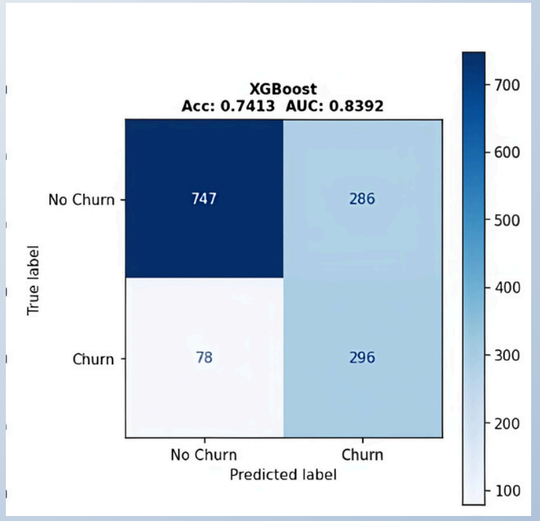
    accuracy                           0.76      1407
   macro avg       0.71      0.76      0.72      1407
weighted avg       0.80      0.76      0.77      1407

==========================================================
```

Model Training and Evaluation



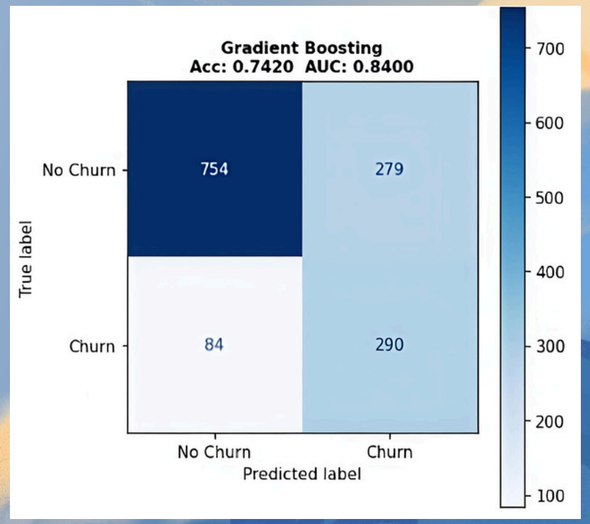
XG Boost



XGBoost

	precision	recall	f1-score	support
No Churn	0.91	0.72	0.80	1033
Churn	0.51	0.79	0.62	374
accuracy	0.74		0.74	1407
macro avg	0.71	0.76	0.71	1407
weighted avg	0.80	0.74	0.76	1407
	0.74	0.74	0.76	1407

Gradient Boosting



Classification Report – Gradient Boosting

	precision	recall	f1-score	support
No Churn	0.90	0.73	0.81	1033
Churn	0.51	0.78	0.62	374
accuracy			0.74	1407
macro avg	0.70	0.75	0.71	1407
weighted avg	0.80	0.74	0.76	1407

Comparison Analysis

Accuracy Ranking (Best → Worst)
Random Forest — 0.756
Gradient Boosting — 0.742
XGBoost — 0.741
Logistic Regression — 0.732

Best Accuracy → Random Forest
Best Churn Detection / Class Separation → Gradient Boosting
Most Stable Overall → Gradient Boosting / XGBoost
Most Interpretable → Logistic Regression

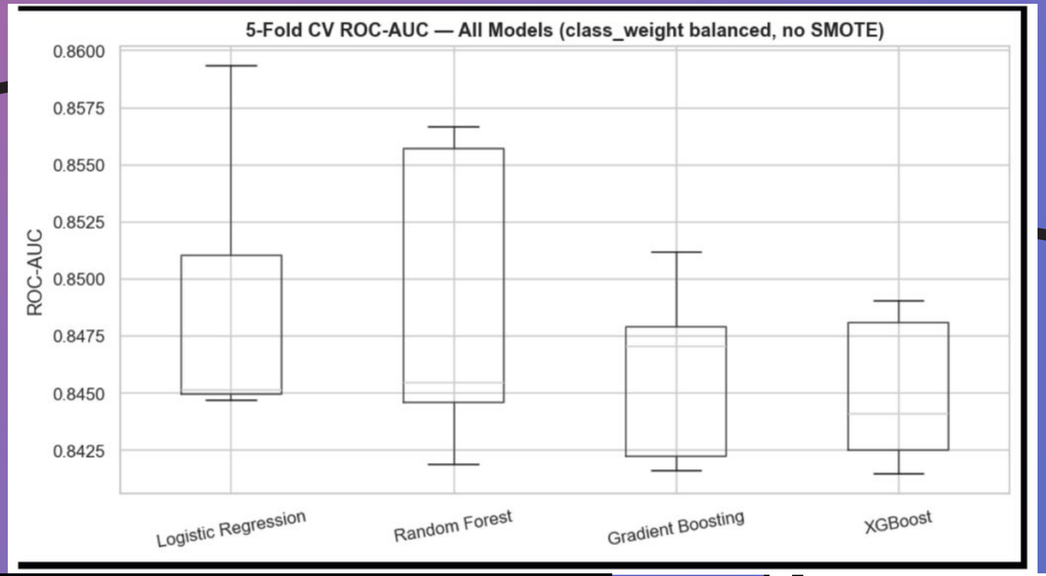
ROC-AUC (Class Separation Ability)
Gradient Boosting — 0.840 (Best)
XGBoost — 0.839
Logistic Regression — 0.837
Random Forest — 0.837
All models show strong discrimination (~0.84) → good churn prediction capability.

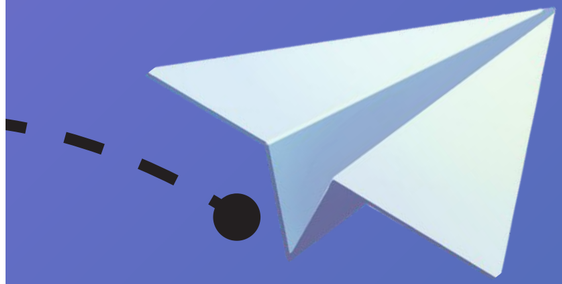
All models perform closely (0.73–0.76 accuracy range). Only ~2–3% difference between best and worst → problem moderately separable.

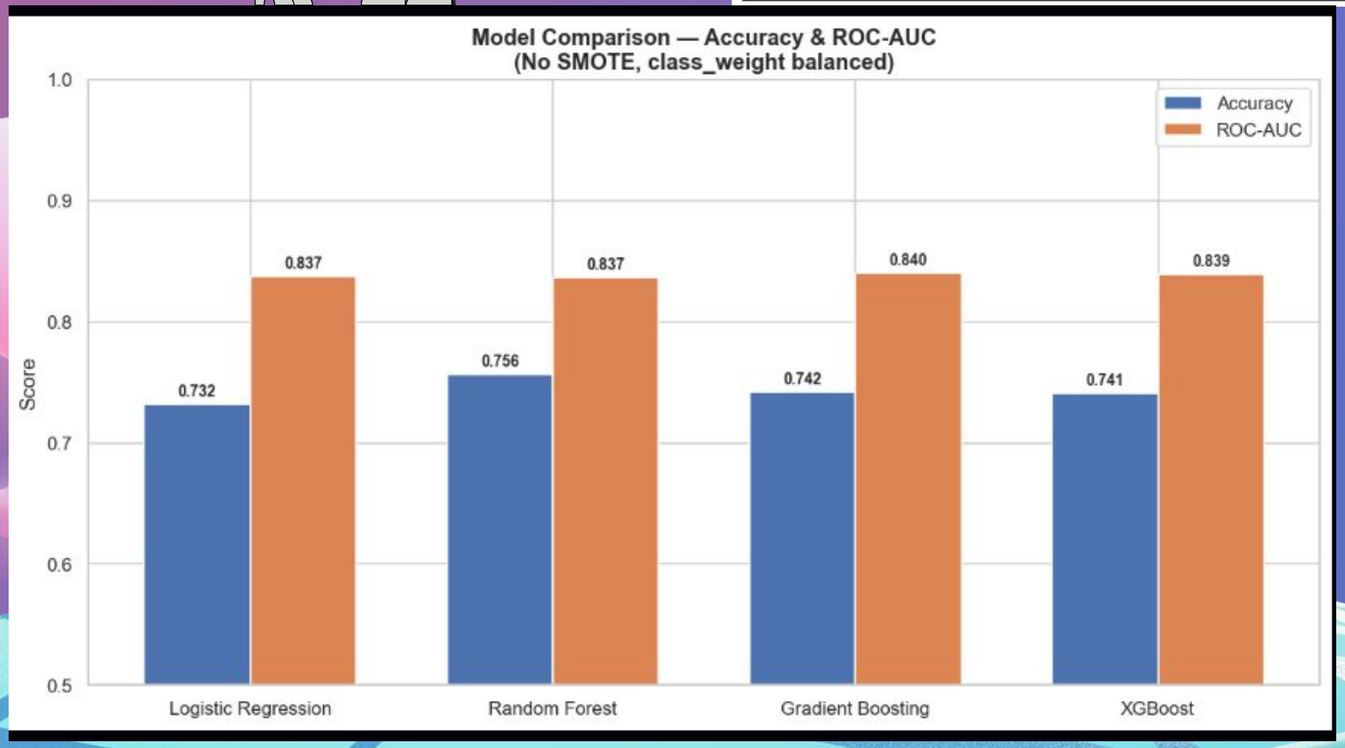


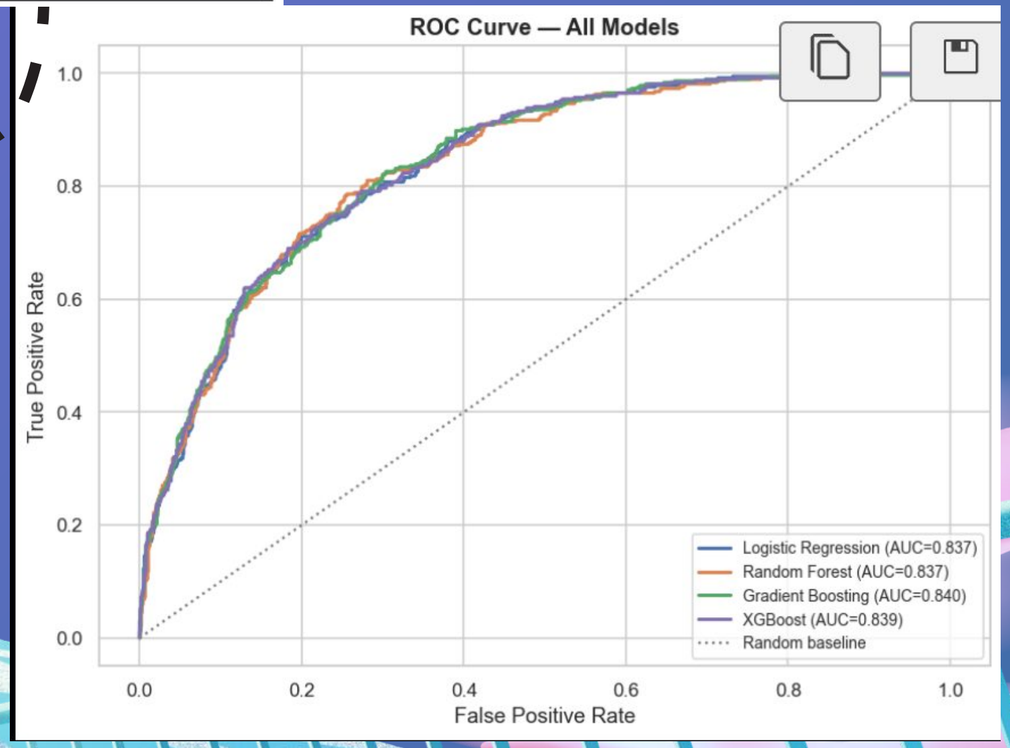
Comparison Analysis











Challenges



- During hyperparameter tuning, the model's accuracy started decreasing because tuning often reduces overfitting and forces the model to generalize better on unseen data. Earlier, the model might have memorized training patterns, giving high accuracy, but after tuning parameters such as max depth, learning rate, or number of trees, the model became more balanced and realistic, which can lower raw accuracy while improving true predictive quality.

- In customer churn prediction, maintaining class balance is crucial because churned customers usually form a small portion of the dataset. If the classes are imbalanced, the model becomes biased toward predicting non-churn customers, leading to misleadingly high accuracy. Focusing on balanced class distribution and using metrics like Recall, F1-score, and ROC-AUC ensures the model correctly identifies actual churners and performs reliably in real-world scenarios.

Thank You

